THE 5 PRACTICES for ORCHESTRATING PRODUCTIVE MATHEMATICS DISCUSSIONS

Intentional teaching moves that turn student thinking into meaningful mathematical learning.

1 ANTICIPATING

Predict what students will do and what solution paths or misconceptions they may have when solving a rich mathematical task.

Possible Strategies

3 × 4

Misconceptions

3 + 4 = 7

4 × 3 = 7

2 MONITORING

Circulate during student work to observe strategies and record who is doing what.

Noticings:

- using arrays
- repeated addition
- skip counting
- partial sums
- misconception: 3 + 4

3 SELECTING

Choose specific students to share their work based on the mathematical ideas their strategies bring to light.

3+3+3+3 = 12

4 × 3 = 12

0 4 8 12

4 SEQUENCING

Order the chosen student presentations purposefully (e.g., from a common or naive strategy to a more efficient or abstract one) to maximize learning.

3+3+3+3 = 12	4 × 3 = 12	0 4 8 12
Build on familiar ideas	Make connections	Highlight more efficient strategies

5 CONNECTING

Help the whole class make mathematical connections between different student strategies and underlying big ideas.

Repeated Addition	Arrays	Number Line
3+3+3+3 = 12	4 × 3 = 12	0 4 8 12

Different strategies. Same mathematical idea!

Students' Ideas

Teacher Moves

Deeper Understanding

From student thinking to meaningful mathematical discussions.

Adapted from Smith, M. S., & Stein, M. K. (2011). *5 Practices for Orchestrating Productive Mathematics Discussions.*
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